

Energy-Efficient Distributed Intrusion Detection Leveraging Graph-Smoothness and Sparsity in Consumer IoT WSNs

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Abstract—An effective security approach is essential for wireless sensor networks (WSNs) against various security threats, which helps to estimate the lifetime of the network. This makes it highly valuable to design an Intrusion Detection System (IDS) that consumes less energy and identifies malicious activity precisely. Traditional IDS techniques are limited to detecting only certain types of attacks and often suffer from reduced network longevity. Intrusion detection plays a critical role in consumer electronics by safeguarding user privacy, ensuring reliable device operation, and protecting interconnected smart environments from increasingly sophisticated cyber threats. This paper proposed a Distributed ADMM-based IDS (D-ADMM-IDS) using graph-smoothness priors and sparsity-driven anomaly decomposition for real-time threat detection. The proposed framework optimizes locally at each sensor, exchanging only compressed dual variables, thereby reducing communication by 11.1% compared to existing schemes. The system adopted a convex composite objective, ensuring convergence within 25 ADMM iterations, with residuals below 2.3×10^{-3} and 2.0×10^{-3} . The simulations show that D-ADMM-IDS achieves a detection accuracy of 96.4% while maintaining a false positive rate of only 1.9%. Additionally, the proposed method reduces average detection latency 3.0 s and energy consumption 0.875 J per node.

Index Terms—Consumer IoT, Energy, Intrusion, Graph smoothness, Signal Sparsity.

I. INTRODUCTION

WSNs is a key technology for enabling smart environments, supporting applications like environmental monitoring, precision agriculture, industrial control, intelligent transportation, and critical infrastructure surveillance [1]. These networks comprise many sensor nodes deployed on an area to monitor physical phenomena and communicate via multi-hop transmission. The WSNs face limitations due to their constrained energy, memory, bandwidth, and computational capabilities. WSNs are vulnerable to various cyber and physical threats, such as sinkhole, selective forwarding, Sybil, and false data attacks [2]–[4].

The integration of WSNs into IoT and emerging 6G networks increased the system scalability and complexity. Smart homes, wearables, industrial systems, and transportation all depend on WSN data, expanding the attack surface and

exposing networks to advanced threats. Existing signature-based systems require frequent updates and miss new attacks, while anomaly-based methods are too resource-heavy for sensor nodes. Centralized detection creates high communication overhead and single points of failure, whereas distributed systems often lack coordination and resilience to handle sophisticated threats [5], [6]. In consumer IoT scenarios, these failures can directly threaten user privacy, trust, and safety, making lightweight and intelligent intrusion detection vital for safeguarding devices and applications. Therefore, an IDS framework that is accurate, resource-efficient, scalable, and capable of real-time operation under limited resources is urgently needed [7], [8]. As consumer electronics continue to increase at an extremely high rate, including smart home, wearables, and connected appliances, wireless sensor networks are now firmly rooted in everyday life. Due to compromising performance of the network, security breaches directly jeopardize the privacy, safety, and trust of the user. Conventional intrusion detection systems tend to be too resource-intensive or centralized to work with consumer electronics. This inspires the necessity of a lightweight, distributed, and energy-efficient IDS framework that would be reliable to work within the limited environments of consumer IoT-enabled WSNs.

In this study, a novel Distributed IDS for WSNs in smart consumer environments, utilizing the Alternating Direction Method of Multipliers (ADMM) is proposed. The core innovation is framing intrusion detection as a structured optimization problem that adopted two key properties of WSN communication in consumer networks, including (i) low-rank patterns in typical network traffic caused by spatial and temporal correlations and (ii) sparsity of anomalies, indicating that attacks are localized and rare. By integrating graph-smoothness priors based on the network topology, the system ensures consistency with physical connectivity while identifying attack-related disruptions. The main contributions of this work are:

- Development of an accurate, topology-aware, distributed IDS framework designed specifically for the operational realities of consumer IoT WSNs.
- Formulation of IDS as a graph-smoothness and sparsity-regularized optimization, enabling accurate separation of network behavior from localized attack patterns.
- Design of a novel low-rank and sparse decomposition model, a distributed ADMM-based solution that combined with graph regularization to detect both overall patterns and localized anomalies.
- Mathematical formulation of a grounded optimization algorithm that ensures high detection robustness while reducing energy and bandwidth use.

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By combining distributed optimization, graph-theoretic modeling, and sparse signal recovery, the proposed system enhances the latest capabilities in consumer IoT WSN intrusion detection. The rest of the paper is organized as follows. Related work is presented in Section II. Section III introduces the system model, threat model, including the network overview, communication flow, energy model, and threat model. The proposed intrusion detection mechanism is described in Section IV. Section V presents the analytical analysis of the proposed method, followed by results and discussion in Section VI. Finally, Section VII concludes the paper.

II. RELATED WORK

In this section various methods for IDSs and secure data transmission in WSNs are discussed and summarized in Table I. In [9], the author introduces a lightweight, energy-efficient IDS specifically designed for LoRaWAN end devices, aiming to counteract energy depletion attacks that gradually drain battery life. It tracks sensor energy consumption patterns through distance-metric-based anomaly detection, and used for resource-limited IoT devices and demonstrates high detection accuracy, with a F1 scores above 0.95 with 0.1 and 4.1% energy consumption. A FL framework for intrusion detection in WSNs, combining FedAvg with XGBoost and utilizing fog computing to extend the network edge is presented in [10]. The method improves XGBoost with an enhanced histogram-based approximation tailored for the imbalanced data typical of WSNs and further reduces communication costs using a TOP-K gradient selection to limit uploaded parameters. Experiments show that this approach achieves better detection accuracy and lower energy use compared to traditional methods. The author in [11] proposed an IDS for sensor networks using an enhanced deep neural network (IDNN). The system architecture is optimized with coyote optimization algorithm integrated with global search (COA-GS). The system testing on KDDCup-99 and WSN-DS datasets shows robust performance, achieving approximately 95% accuracy, 96% recall, 95% F1-score, and an ROC AUC around 98%, along with low detection latency (1 ms) and delay (0.8 ms).

A novel FIDS-HMM technique for WSNs is proposed with the integration of fuzzy logic and the hidden Markov model to identify black hole attacks and malicious nodes [13]. The HMM monitors the energy level of the node, while fuzzy inference leverages metrics such as packet arrival rates, packet drop rate, and remaining energy to assess the likelihood of an attack. The simulations were performed in NS-2 which shows that FIDS-HMM enhances QoS metrics—namely packet delivery ratio, throughput, and delay. [13] introduces a collaborative IDS (CIDS) designed for RPL-based networks within the Internet of Everything that focuses on complex joint routing attacks that impair performance and energy efficiency. It integrates queue and workload-aware RPL (QWL-RPL) for congestion control, genetic algorithm-optimized weighted random forwarding for load balancing, fuzzy logic for assessing trust, LightGBM for attack detection, and Q-learning with trickle-time for collaboratively classifying and managing attacks. The results show improved detection accuracy, higher throughput,

and better energy efficiency, along with lower control overhead and faster computation. Author in [14] presents TEAD, a hybrid IDS designed for IoT-enabled WSNs. It combines trust assessment with ML-based anomaly detection. TEAD calculates real-time trust scores for nodes using behavioral indicators like communication link reliability, weight ratio, and frequency deviation. Testing on the NSL-KDD and CIC-IDS-2017 datasets shows TEAD's high effectiveness, with 96% accuracy, a 2% false positive rate, and a 3% false negative rate, making it well-suited for resource-limited IoT-WSNs. Another study introduces an energy-efficient data transmission framework for WSNs that combines the Low-Energy Adaptive Clustering Hierarchy (LEACH) protocol with a fuzzy-based IDS [15]. LEACH reduces energy use by randomly selecting cluster heads (CHs) and rotating the role among nodes, promoting balanced energy consumption. The Fuzzy-IDS uses fuzzy logic and artificial neural networks (ANNs) to identify nodes as either legitimate or malicious, strengthening network security. Simulation results show that the proposed method achieves about 97% accuracy, precision, and specificity, with low false positive and negative rates, thereby enhancing data transmission efficiency and extending network lifespan.

III. SYSTEM MODEL

This section discusses the architecture of consumer WSNs and presents a threat model that identifies intrusions based on a formal mathematical framework for precise analysis.

A. Smart Environment and Network Overview

A smart environment can be described as a physical facility, e.g., a smart building, agricultural field, or urban infrastructure, integrated with ubiquitous sensing, computation, and communication resources. In our model, this environment is equipped with a set of N wireless sensor nodes, represented by:

$$\mathcal{V} = \{v_1, v_2, \dots, v_N\}. \quad (1)$$

Each node v_i consists of sensing modalities, a microcontroller, a radio transceiver, and a limited-capacity power source. The consumer network operates in a *multi-hop* topology, in which is data originating at a source node may traverse intermediate nodes before reaching the sink or base station. Let $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ represent the set of bidirectional communication links. The network can be represented as an undirected graph:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}). \quad (2)$$

Nodes $(v_i, v_j) \in \mathcal{E}$, communicate directly if their Euclidean distance satisfies $d_{ij} \leq R_c$, where communication range R_c is determined by the radio hardware and path-loss conditions. In the presence of node mobility or topology changes, the graph Laplacian is updated dynamically. As long as topology variations occur on a slower timescale than the ADMM iterations; the graph-smoothness regularization remains effective and preserves convergence properties.

B. Communication and Data Flow Model

Time is expressed as discrete slots $t \in \mathbb{N}^+$. In each time slot, a node may be in *sense*, *transmit*, *receive*, or *idle* status.

TABLE I: Summary of Intrusion Detection Approaches in WSNs and IoT Networks

Ref	Target Network	Methodology	Main Contribution / Focus	Performance Highlights
9	LoRaWAN End Devices	Lightweight IDS, energy consumption monitoring, anomaly detection	Detects energy depletion attacks efficiently for resource-constrained IoT devices	High detection performance with low resource
10	WSNs	FedAvg (federated learning), XGBoost, TOP-K gradient selection	Federated IDS for WSNs to detect intrusions while reducing communication overhead	Improved detection accuracy and energy efficiency
11	WSNs	Deep Neural Network (DNN), Coyote Optimization Algorithm (COA-GS)	Optimized IDS design for WSNs with enhanced detection	~95% accuracy, low detection latency (~1 ms)
12	WSNs	Fuzzy logic, Hidden Markov Model (HMM)	Effective detection of black hole attacks in WSNs	Enhanced packet delivery ratio, throughput, and delay
13	RPL-based IoE Networks	Collaborative IDS, QWL-RPL, fuzzy logic, LightGBM, Q-learning	Detects joint routing attacks in low-power networks	Higher accuracy, throughput, and energy efficiency
14	IoT-enabled WSNs	Trust evaluation, LSTM, autoencoder anomaly detection	TEAD framework combining trust metrics with ML-based IDS	96% accuracy, 2% FPR, 3% FNR
15	WSNs	LEACH clustering, Fuzzy logic, ANN IDS	Energy-efficient data transmission with fuzzy IDS routing	~97% accuracy, precision, and specificity, low false positive/negative

The sensed value at node v_i and time t is denoted by:

$$s_i(t) \in \mathbb{R}. \quad (3)$$

The data packet transmitted from v_i to v_j is denoted by $p_{ij}(t)$, that contains the sensed value and optional metadata like timestamp and node ID.

$$\mathcal{N}(i) = \{v_j \in \mathcal{V} \mid (v_i, v_j) \in \mathcal{E}\}. \quad (4)$$

The network adopted a store-and-forward routing paradigm. Let $\mathcal{P}_{i \rightarrow \text{sink}}$ represent a path from node v_i to the sink node v_{sink} . The *path length* is the number of hops:

$$L(\mathcal{P}_{i \rightarrow \text{sink}}) = \text{cardinality of edges in path}. \quad (5)$$

The *end-to-end delay* for a packet from v_i to the sink is:

$$D_i = \sum_{(v_m, v_n) \in \mathcal{P}_{i \rightarrow \text{sink}}} \left(T_{\text{tx}}^{(m,n)} + T_{\text{queue}}^{(m)} + T_{\text{proc}}^{(m)} \right), \quad (6)$$

where $T_{\text{tx}}^{(m,n)}$ is the transmission delay between v_m and v_n , $T_{\text{queue}}^{(m)}$, $T_{\text{proc}}^{(m)}$ is the queuing and processing delay.

C. Energy Model

Energy efficiency is essential for the networks because nodes typically run on batteries. The energy used to transmit data from v_i to v_j is described by the first-order radio model.

$$E_{\text{tx}}(k, d_{ij}) = E_{\text{elec}} \cdot k + E_{\text{amp}} \cdot k \cdot d_{ij}^{\gamma}, \quad (7)$$

where k is a packet size in bits, E_{elec} is the energy per bit, E_{amp} is the energy per bit per m^{γ} , and γ is a path-loss exponent. The energy consumption for receiving a k -bit packet is:

$$E_{\text{rx}}(k) = E_{\text{elec}} \cdot k. \quad (8)$$

The energy update equation for a node v_i is therefore:

$$E_i(t+1) = E_i(t) - E_{\text{tx}}^{(i)}(t) - E_{\text{rx}}^{(i)}(t) - E_{\text{proc}}^{(i)}(t), \quad (9)$$

where $E_{\text{proc}}^{(i)}(t)$ is the energy spent on computation and sensing.

D. Channel Model

The wireless channel connecting two nodes experiences fading and noise. For heterogeneous sensor, the measurement noise variance is either obtained from device specifications or estimated empirically during a short calibration phase. Each node maintains a local noise estimate, allowing the model to accommodate sensor-dependent uncertainty without requiring global synchronization. We represent the baseband equivalent channel coefficient between v_i and v_j at time t as follows:

$$h_{ij}(t) = \sqrt{G_{ij}} \cdot \eta_{ij}(t), \quad (10)$$

where, G_{ij} is a large-scale path gain, inversely proportional to d_{ij}^{γ} , $\eta_{ij}(t)$ is a small-scale fading coefficient (Rayleigh fading). The received signal at v_j from v_i is:

$$r_j(t) = h_{ij}(t) \cdot x_i(t) + n_j(t), \quad (11)$$

where $x_i(t)$ is the transmitted symbol and $n_j(t)$ is additive white Gaussian noise with variance σ_n^2 .

E. Threat Model

In a smart WSNs, an adversary aims to disrupt or manipulate sensing, communication, or decision-making. The adversary may be external, injecting, jamming, or eavesdropping on traffic, or internal, having compromised nodes with access to keys, memory, or sensing functions. These attackers can intercept or modify messages, or tamper with node hardware or firmware. Let $y_i(t)$ be the true sensed value at node v_i . Under attack, the observed value $\tilde{y}_i(t)$ can be modeled as

$$\tilde{y}_i(t) = y_i(t) + a_i(t), \quad (12)$$

where $a_i(t)$ is the disturbance caused by the adversary. The resulting packet delivery ratio (PDR) is

$$\text{PDR}_{\text{attack}} = \frac{\sum_t \sum_i (1 - P_{\text{loss}}^{(i)}(t))}{\sum_t N}. \quad (13)$$

Assuming that the network topology, represented as $\mathcal{G}$, remains connected under normal conditions because of a trusted, physically secure sink node. Also, the time synchronization is accurate for slot-based modeling, and that communication occurs over half-duplex shared channels. The ADMM penalty parameter ρ is important for governing the balance between

primal consistency and dual update magnitude. As ADMM ensures convergence for any $\rho > 0$ under convexity, the practical convergence speed is highly sensitive to its value. A small value leads to slow enforcement, whereas an excessively large one ρ may cause oscillatory updates or numerical instability.

IV. PROPOSED INTRUSION DETECTION MECHANISM

This section describes a detailed, mathematical formulation and description of the proposed IDS designed for consumer IoT WSNs. Consider a network with N sensors, where the connectivity is depicted by an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. At each discrete time index $t \in \{1, 2, \dots, T\}$ the sensors gather a vector of scalar measurements denoted by,

$$\mathbf{y}(t) = [y_1(t), y_2(t), \dots, y_N(t)]^\top \in \mathbb{R}^N. \quad (14)$$

These measurements result from the physical process, measurement noise, and possible malicious perturbations. The observed vector can be conceptually separated as

$$\mathbf{y}(t) = \mathbf{s}(t) + \mathbf{a}(t) + \mathbf{n}(t), \quad (15)$$

where $\mathbf{s}(t)$ the legitimate signal component is, $\mathbf{a}(t)$ indicate an adversarial perturbation, and $\mathbf{n}(t)$ is the stochastic measurement noise. The $\mathbf{s}(t)$ is assumed to change rapidly over the graph. The adversarial component $\mathbf{a}(t)$ is defined as spatially sparse and possibly temporally sporadic. To formalize smoothness in the network, let $\mathbf{L} \in \mathbb{R}^{N \times N}$ is the combinatorial graph Laplacian integrated with $\mathcal{G}$, defined by $\mathbf{L} = \mathbf{D} - \mathbf{W}$ where $\mathbf{W}$ is the symmetric weighted adjacency matrix and $\mathbf{D} = \text{diag}(d_1, \dots, d_N)$ is the diagonal degree matrix with $d_i = \sum_j W_{ij}$. The graph Laplacian encodes local differences between node values, for any vector $\mathbf{u} \in \mathbb{R}^N$ the quadratic form $\mathbf{u}^\top \mathbf{L} \mathbf{u} = \frac{1}{2} \sum_{i,j} W_{ij} (u_i - u_j)^2$ computes the weighted variation of $\mathbf{u}$ on the graph. Given a temporal window $t = 1, \dots, T$ we obtained the measurements into a matrix

$$\mathbf{Y} = [\mathbf{y}(1), \mathbf{y}(2), \dots, \mathbf{y}(T)] \in \mathbb{R}^{N \times T}. \quad (16)$$

Similarly, we define the matrices for legitimate signals, adversarial perturbations, and noise as $\mathbf{S}$, $\mathbf{A}$ and $\mathbf{N}$ respectively, so that a concise model for the observation window becomes:

$$\mathbf{Y} = \mathbf{S} + \mathbf{A} + \mathbf{N}. \quad (17)$$

The detection task is to identify nonzero entries of $\mathbf{A}$, including their spatial and temporal support, and measure their magnitudes, while estimating the smooth component $\mathbf{S}$.

$$\min_{\mathbf{S}, \mathbf{A}} \frac{1}{2} \|\mathbf{Y} - \mathbf{S} - \mathbf{A}\|_F^2 + \alpha \text{tr}(\mathbf{S}^\top \mathbf{L} \mathbf{S}) + \beta \|\mathbf{A}\|_{1,1} + \gamma \mathcal{R}_t(\mathbf{S}). \quad (18)$$

where $\frac{1}{2} \|\mathbf{Y} - \mathbf{S} - \mathbf{A}\|_F^2$ ensures that the data remains accurate even in noisy environments. Meanwhile, the Laplacian quadratic $\text{tr}(\mathbf{S}^\top \mathbf{L} \mathbf{S}) = \sum_{t=1}^T \mathbf{s}(t)^\top \mathbf{L} \mathbf{s}(t) = \frac{1}{2} \sum_t \sum_{i,j} W_{ij} (s_i(t) - s_j(t))^2$ encourages spatial smoothness by penalizing ℓ_1 norm $\|\mathbf{A}\|_{1,1} = \sum_{i,t} |A_{i,t}|$ promotes sparsity in the adversarial matrix, which aligns with the idea that attacks typically impact only a limited number of sensors. The temporal regularizer $\mathcal{R}_t(\mathbf{S})$ ensures that $\mathbf{S}$ captures the slow dynamics of the legitimate process. The proposed model uses

a combination of both time regularization and space Laplacian coupling to retrieve the natural framework of intrusion conduct in the WSNs. Temporal regularization imposes smoothness of the legitimate signal through the consecutive time indices and guarantees smooth dynamics of the system that vary gradually, and abrupt deviations, which are usually brought about by intrusions, are isolated in the sparse component. At the same time, spatial Laplacian prior utilizes the similarities between adjacent nodes: legitimate behavior is more likely to develop in a coherent manner throughout the graph, but attacks are typically local or asymmetric. The temporal regularizer $\mathcal{R}_t(\mathbf{S})$ refers to the mathematical constraint added to stabilize and smooth the sparse intrusion signal $\mathbf{S}$ across time.

$$\mathcal{R}_t(\mathbf{S}) = \sum_{t=2}^T \|\mathbf{s}(t) - \mathbf{s}(t-1)\|_2^2 = \|\mathbf{H}\mathbf{S}\|_F^2, \quad (19)$$

where, $\mathbf{H} \in \mathbb{R}^{T \times (T-1)}$ is the first-difference operator, where the entries are defined as $H_{t,t} = 1$, $H_{t+1,t} = -1$. The optimization problem in equation (18) is convex and amenable to powerful numerical methods. However, the centralization implied by solving (18) in a single location is impractical for large-scale WSNs. Let $\mathbf{s}_i \in \mathbb{R}^T$ is the temporal sequence of legitimate values at node i , and $\mathbf{a}_i \in \mathbb{R}^T$ represent the temporal attack sequence at node i . The matrices can be decomposed as $\mathbf{S} = [\mathbf{s}_1^\top; \dots; \mathbf{s}_N^\top]$ and $\mathbf{A} = [\mathbf{a}_1^\top; \dots; \mathbf{a}_N^\top]$. The Laplacian smoothness can be expressed using the edge incidence.

$$\text{tr}(\mathbf{S}^\top \mathbf{L} \mathbf{S}) = \frac{1}{2} \sum_{(i,j) \in \mathcal{E}} W_{ij} \|\mathbf{s}_i - \mathbf{s}_j\|_2^2. \quad (20)$$

The Frobenius data-fidelity term decomposes as a sum of nodewise squared errors $\frac{1}{2} \sum_{i=1}^N \|\mathbf{y}_i - \mathbf{s}_i - \mathbf{a}_i\|_2^2$ where $\mathbf{y}_i$ denotes the i th row of $\mathbf{Y}$ and similarly for the other matrices. For each node i we introduce local copies of its own signal and attack variables. More specifically, define for each edge $(i, j) \in \mathcal{E}$ an auxiliary variable $\mathbf{z}_{ij} \in \mathbb{R}^T$ intended to represent the local agreement between $\mathbf{s}_i$ and $\mathbf{s}_j$. The constraints satisfy that $\mathbf{z}_{ij} = \mathbf{s}_i$ and $\mathbf{z}_{ji} = \mathbf{s}_j$ in the consensus limit. Using these auxiliary variables, the optimization can be equivalently expressed (for the squared-difference temporal regularizer) as

$$\begin{aligned} \min_{\{\mathbf{s}_i, \mathbf{a}_i\}, \{\mathbf{z}_{ij}\}} & \frac{1}{2} \sum_{i=1}^N \|\mathbf{y}_i - \mathbf{s}_i - \mathbf{a}_i\|_2^2 + \alpha \frac{1}{2} \sum_{(i,j) \in \mathcal{E}} W_{ij} \|\mathbf{z}_{ij} - \mathbf{z}_{ji}\|_2^2 \\ & + \beta \sum_{i=1}^N \|\mathbf{a}_i\|_1 + \gamma \sum_{i=1}^N \|\mathbf{D}_t \mathbf{s}_i\|_2^2 \end{aligned} \quad (21)$$

subject to $\mathbf{z}_{ij} = \mathbf{s}_i$, $\mathbf{z}_{ji} = \mathbf{s}_j$, $\forall (i, j) \in \mathcal{E}$. $\mathbf{D}_t$ is the temporal-difference operator such that $\mathbf{D}_t \mathbf{s}_i$ yields the first differences of $\mathbf{s}_i$. The augmented Lagrangian for the constrained problem (21) in Lagrange multipliers $\{\lambda_{ij}, \lambda_{ji}\}$ that correspond to the constraints $\mathbf{z}_{ij} - \mathbf{s}_i = \mathbf{0}$ and $\mathbf{z}_{ji} - \mathbf{s}_j = \mathbf{0}$,

respectively. The resulting Lagrangian is:

$$\begin{aligned} \mathcal{L}_\rho = & \frac{1}{2} \sum_{i=1}^N \|\mathbf{y}_i - \mathbf{s}_i - \mathbf{a}_i\|_2^2 + \alpha \frac{1}{2} \sum_{(i,j) \in \mathcal{E}} W_{ij} \|\mathbf{z}_{ij} - \mathbf{z}_{ji}\|_2^2 \\ & + \beta \sum_{i=1}^N \|\mathbf{a}_i\|_1 + \gamma \sum_{i=1}^N \|\mathbf{D}_t \mathbf{s}_i\|_2^2 \\ & + \sum_{(i,j) \in \mathcal{E}} \langle \boldsymbol{\lambda}_{ij}, \mathbf{z}_{ij} - \mathbf{s}_i \rangle + \sum_{(i,j) \in \mathcal{E}} \frac{\rho}{2} \|\mathbf{z}_{ij} - \mathbf{s}_i\|_2^2. \end{aligned} \quad (22)$$

The ADMM proceeds by alternating minimization over the primal blocks $\{\mathbf{s}_i\}$, $\{\mathbf{a}_i\}$, and $\{\mathbf{z}_{ij}\}$ followed by dual updates for the multipliers. At ADMM iteration k the $\mathbf{a}_i$ update solves:

$$\mathbf{a}_i^{(k+1)} = \arg \min_{\mathbf{a}_i} \frac{1}{2} \|\mathbf{y}_i - \mathbf{s}_i^{(k)} - \mathbf{a}_i\|_2^2 + \beta \|\mathbf{a}_i\|_1. \quad (23)$$

The residual vector $\mathbf{r}_i^{(k)} = \mathbf{y}_i - \mathbf{s}_i^{(k)}$, the solution is given by soft-thresholding:

$$\mathbf{a}_i^{(k+1)} = \mathcal{S}_\beta(\mathbf{r}_i^{(k)}), \quad (24)$$

The $\mathbf{z}_{ij}$ -updates at iteration k solve, for every edge $(i, j) \in \mathcal{E}$,

$$\begin{aligned} \mathbf{z}_{ij}^{(k+1)} = & \arg \min_{\mathbf{z}_{ij}} \alpha W_{ij} \|\mathbf{z}_{ij} - \mathbf{z}_{ji}^{(k)}\|_2^2 \\ & + \langle \boldsymbol{\lambda}_{ij}^{(k)}, \mathbf{z}_{ij} - \mathbf{s}_i^{(k)} \rangle + \frac{\rho}{2} \|\mathbf{z}_{ij} - \mathbf{s}_i^{(k)}\|_2^2. \end{aligned} \quad (25)$$

This is a quadratic problem and yields a closed-form linear update, which is rewritten as:

$$\mathbf{z}_{ij}^{(k+1)} = (2\alpha W_{ij} \mathbf{I} + \rho \mathbf{I})^{-1} (2\alpha W_{ij} \mathbf{z}_{ji}^{(k)} + \rho \mathbf{s}_i^{(k)} - \boldsymbol{\lambda}_{ij}^{(k)}). \quad (26)$$

The $\mathbf{s}_i$ is computationally involved because it integrates data fidelity, temporal smoothness, and the consensus terms related to incident edges. The node update is solve by:

$$\begin{aligned} \mathbf{s}_i^{(k+1)} = & \arg \min_{\mathbf{s}_i} \frac{1}{2} \|\mathbf{y}_i - \mathbf{s}_i - \mathbf{a}_i^{(k+1)}\|_2^2 + \gamma \|\mathbf{D}_t \mathbf{s}_i\|_2^2 \\ & + \sum_{j \in \mathcal{N}(i)} \left(-\langle \boldsymbol{\lambda}_{ij}^{(k)}, \mathbf{s}_i \rangle + \frac{\rho}{2} \|\mathbf{z}_{ij}^{(k+1)} - \mathbf{s}_i\|_2^2 \right). \end{aligned} \quad (27)$$

Since all terms are quadratic, $\mathbf{s}_i$ minimization simplifies to solving a linear system. Gathering the quadratic and linear components results in a system expressed as:

$$\mathbf{Q}_i \mathbf{s}_i^{(k+1)} = \mathbf{b}_i^{(k)}, \quad (28)$$

where the matrix $\mathbf{Q}_i \in \mathbb{R}^{T \times T}$ is positive definite. Solving the local update for $\mathbf{s}_i$ requires $O(T)$ operations using the banded structure of $\mathbf{Q}_i$, while the soft-thresholding update for $\mathbf{a}_i$ is $O(T)$ per node, making each ADMM iteration computationally lightweight for WSN hardware.

$$\mathbf{Q}_i = \mathbf{I}_T + 2\gamma \mathbf{D}_t^\top \mathbf{D}_t + \rho |\mathcal{N}(i)| \mathbf{I}_T, \quad (29)$$

$$\mathbf{b}_i^{(k)} = \mathbf{y}_i - \mathbf{a}_i^{(k+1)} + \sum_{j \in \mathcal{N}(i)} (\rho \mathbf{z}_{ij}^{(k+1)} - \boldsymbol{\lambda}_{ij}^{(k)}). \quad (30)$$

Solving this linear system provides the current legitimate signal for node i . When $\mathbf{Q}_i$ is banded, which occurs if $\mathbf{D}_t$ is the first-difference operator, efficient banded or Cholesky

solvers can be used. The temporal window T manages the trade-off between detection delay and computational effort. Additionally, precomputing a Cholesky factor of $\mathbf{Q}_i$ helps minimize runtime on platforms with limited resources. The dual variables are updated in the usual ADMM fashion:

$$\boldsymbol{\lambda}_{ij}^{(k+1)} = \boldsymbol{\lambda}_{ij}^{(k)} + \rho (\mathbf{z}_{ij}^{(k+1)} - \mathbf{s}_i^{(k+1)}), \quad \forall (i, j) \in \mathcal{E}. \quad (31)$$

These dual updates are distributed, matching the primal steps locality, nodes exchange their $\mathbf{s}_i$ and $\mathbf{z}_{ij}$ values with neighbors for updates. Communication is reduced through lightweight compression methods like quantization or sketching. ADMM maintains global convergence and its speed improves with proper scaling of W_{ij} and warm-starting across windows. The false alarm control is controlled by statistically calibrated thresholds $\tau = \sigma_r \Phi^{-1}(1 - \alpha_f/2)$, with asynchronous spatially adaptive variants used for heterogeneous sensor noise. Consider the statistic for node i at time t be:

$$T_{i,t} = \sum_{\tau=t-M+1}^t \log \frac{f_1(\hat{A}_{i,\tau})}{f_0(\hat{A}_{i,\tau})}, \quad (32)$$

where f_1 and f_0 are the likelihood functions. The proposed method enhances sensor network security by integrating smoothness and sparsity models. Algorithm 1 focuses on processing at the local node level. Each sensor node independently monitors its environment, filters data, and extracts features like sudden changes, noise, or patterns. Algorithm 2 employs an ADMM-based consensus update, where nodes share partial information to iteratively distinguish between a smooth, legitimate signal and a sparse anomaly component. ADMM facilitates agreement among nodes with minimal communication, enhancing anomaly detection and noise reduction to differentiate normal fluctuations from genuine intrusion attacks. In Algorithm 3, an intrusion detection rule is applied using anomaly estimates at each node. Instead of arbitrary thresholds, it uses a statistically derived threshold based on residual variance.

Algorithm 1 Local Node Processing at Sensor i

Require: Observations $\mathbf{y}_i(t)$ for $t = 1, \dots, T$

- 1: Preprocess signal: remove mean and normalize $\mathbf{y}_i(t)$
- 2: Extract local features: temporal differences, variance, and correlations
- 3: Compute preliminary anomaly score $s_i(t)$
- 4: Store $s_i(t)$ and prepare local data for ADMM update

Ensure: Feature vector $\mathbf{f}_i(t)$ and anomaly score $s_i(t) = 0$

Algorithm 2 Distributed ADMM Consensus Update

Require: Local observations $\mathbf{y}_i$, preliminary $\mathbf{s}_i$, neighbors $\mathcal{N}_i$

- 1: Initialize $\mathbf{s}_i^{(0)}$, $\mathbf{a}_i^{(0)}$, dual variable $\boldsymbol{\lambda}_i^{(0)}$
- 2: **for** each iteration $k = 1, 2, \dots, K$ **do**
- 3: Update smooth signal $\mathbf{s}_i^{(k+1)}$
- 4: Update anomaly component via soft-thresholding $\mathbf{a}_i^{(k+1)}$
- 5: Update dual variable $\boldsymbol{\lambda}_i^{(k+1)}$
- 6: **end for**

Ensure: Local estimates $\hat{\mathbf{s}}_i$, $\hat{\mathbf{a}}_i = 0$

Algorithm 3 Intrusion Detection and Decision Rule

Require: Estimated anomaly signals $\hat{\mathbf{a}}_i(t)$, residuals $\mathbf{r}_i(t)$

- 1: Estimate residual variance σ_r^2 over calibration window
- 2: Set threshold, τ
- 3: **for** each node i and time t **do**
- 4: **if** $|\hat{A}_{i,t}| > \tau$ **then**
- 5: Declare intrusion at node i , time t
- 6: **else**
- 7: No intrusion
- 8: **end if**
- 9: **end for**

Ensure: Intrusion decision map $\mathcal{D}(i, t) = 0$

Lemma 4.1 (First-order optimality and subgradient characterization): Let $\mathbf{Y}$, $\mathbf{L}$, and the regularization parameters α, β, γ be given. Any pair $(\mathbf{S}^*, \mathbf{A}^*)$ that minimizes

$$\mathcal{J}(\mathbf{S}, \mathbf{A}) = \frac{1}{2} \|\mathbf{Y} - \mathbf{S} - \mathbf{A}\|_F^2 + \alpha \text{tr}(\mathbf{S}^\top \mathbf{L} \mathbf{S}) + \beta \|\mathbf{A}\|_{1,1} + \gamma \mathcal{R}_t(\mathbf{S}) \quad (33)$$

satisfies, for i and time index t , the stationarity conditions

$$\mathbf{0} = -(\mathbf{Y} - \mathbf{S}^* - \mathbf{A}^*) + 2\alpha \mathbf{L} \mathbf{S}^* + \gamma \nabla_{\mathbf{S}} \mathcal{R}_t(\mathbf{S}^*) \quad (34)$$

$$\mathbf{0} \in -(\mathbf{Y} - \mathbf{S}^* - \mathbf{A}^*) + \beta \partial \|\mathbf{A}^*\|_{1,1}, \quad (35)$$

where $\partial \|\cdot\|_{1,1}$ is the subdifferential of the mixed ℓ_1 norm.

Theorem 4.2 (Convergence of the distributed ADMM solver): Assuming that the temporal regularizer $\mathcal{R}_t$ is convex and the central objective is convex and closed, then for fixed penalty parameter $\rho > 0$, the iterates generated by ADMM will converge to a primal-dual optimal $(\mathbf{S}^*, \mathbf{A}^*, \{\mathbf{z}_{ij}^*\}, \{\boldsymbol{\lambda}_{ij}^*\})$ provided the network graph remains connected and each local subproblem is solved exactly at every iteration.

Definition 1 (Low-frequency smooth subspace): Let $\mathbf{L} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top$ be the eigen-decomposition of the Laplacian, and $\mathcal{U}_r$ represents the subspace spanned by the r lowest-frequency eigenvectors, i.e., $\mathcal{U}_r = \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_r\}$. A matrix $\mathbf{S}$ is said to be r -bandlimited in time if each temporal column $\mathbf{s}(t)$ is contained within $\mathcal{U}_r$.

Theorem 4.3 (Sufficient condition for exact recovery in the noiseless case): Suppose the observations that follow the equation $\mathbf{Y} = \mathbf{S}_0 + \mathbf{A}_0$, where $\mathbf{S}_0$ is perfectly r -bandlimited, and $\mathbf{A}_0$ contains no more than s nonzero entries in the $N \times T$ matrix. The incoherence parameter:

$$\mu = \max_{i=1, \dots, N} \max_{k=1, \dots, r} \sqrt{N} |U_{ik}|, \quad (36)$$

measures how concentrated the low-frequency eigenvectors are in the canonical form. If the pair (r, s) satisfies the relation

$$s \leq c \cdot \frac{NT}{\mu^2 r \log(NT)} \quad (37)$$

for a positive numerical constant c .

Theorem 4.4 (Stable recovery under Gaussian noise): Assume $\mathbf{Y} = \mathbf{S}_0 + \mathbf{A}_0 + \mathbf{N}$, where $\mathbf{N}$ consists of independent and identically distributed Gaussian entries with a variance of σ^2 . Given the same incoherence and sparsity, the minimizer $(\hat{\mathbf{S}}, \hat{\mathbf{A}})$ meets the error bound.

$$\|\hat{\mathbf{S}} - \mathbf{S}_0\|_F + \|\hat{\mathbf{A}} - \mathbf{A}_0\|_F \leq C \cdot \sigma \sqrt{NT}, \quad (38)$$

with high probability, where C is a constant that depends polynomially on μ , r , and the chosen regularization parameters.

V. RESULTS AND DISCUSSION

To assess how well the proposed Distributed ADMM-based IDS (D-ADMM-IDS) performs, we set up a detailed array of simulation parameters. The N sensor nodes are deployed randomly across a 2D space measuring $500 \times 500 \text{ m}^2$. The communication radius R_c is kept at a steady 50 m, which helped maintain a connected network with an average node degree of about $\bar{d} \approx 8$. Each node was assumed to use an IEEE 802.15.4-compliant transceiver, with the energy cost per packet κ calculated. The sensing process $x_i(t)$ at each node i and time step t , influenced by Gaussian noise $n_i(t) \sim \mathcal{N}(0, \sigma^2)$ with σ set according to the targeted SNR. The high-amplitude anomalies $a_i(t)$ were embed into a fraction ϕ of nodes ($\phi = 1\% - 30\%$). The ADmm parameters were tuned for fast convergence and low communication overhead, with penalty $\rho = 1.0$ and stopping criterion based on primal and dual residuals below 10^{-4} . The simulation ran for $T = 100$ steps, and the performance was analyzed using accuracy $\mathcal{A}$, false positive rate $\mathcal{F}_P$, accounting for computation and communication costs. The simulation parameters are summarized in Table II. The anomaly model tests the IDS against unusual attack signatures, with attack intensity σ . This demonstrates how D-ADMM-IDS surpasses current methods, comparing it with enhanced graphSAGE-based IDS [16], multi-channel gated recurrent unit [17], and a classical graph-based anomaly detector [18]. The detection accuracy η as a function of the attack intensity ρ_a , where ρ_a is represented as the proportion of compromised nodes to the total network size, $\rho_a = \frac{|\mathcal{N}_{\text{attack}}|}{|\mathcal{N}|}$ is shown in Figure 1a. The accuracy of the proposed D-ADMM-IDS is above 94.8% even at $\rho_a = 0.15$. In comparison, enhanced GraphSAGE-based IDS [16] drops from 92.3% to 85.4%, Multi-Channel Gated Recurrent Unit [17] from 90.1% to 83.0%, and the classical graph-based anomaly detector [18] from 87.9% to 80.5% for same range.

TABLE II: Simulation Parameters for D-ADMM-IDS

Parameter	Value
Number of nodes N	50–300
Deployment area	$500 \times 500 \text{ m}^2$
Communication radius R_c	50 m
Mean node degree $\bar{d}$	≈ 8
Sensing noise standard deviation σ	0.05 (normalized units)
Attack intensity ϕ	0.01–0.30
Anomaly distribution	Laplacian, zero-mean
ADMM penalty parameter ρ	1.0
Per-packet energy cost κ	50 nJ/bit

Figure 1b examines the false positive rate (FPR) as a function of the decision threshold θ , i.e., $\text{FPR}(\theta) = \frac{\text{FP}(\theta)}{\text{FP}(\theta) + \text{TN}(\theta)}$. The D-ADMM-IDS keeps an FPR below 2.8% for all $\theta \geq 0.4$, whereas the enhanced GraphSAGE-based IDS [16] exhibits an FPR as high as 5.5%, the multi-channel gated recurrent unit [17] attains 7.8%, and the classical graph-based anomaly detector [18] exceeds 10.2% for lower thresholds. Interestingly, even with $\theta = 0.7$, our method achieves a false positive rate (FPR) of just 1.9%, which is 3.6%. Figure 1c shows the average energy consumption per node E_n in joules. Comparing it to the Enhanced GraphSAGE-based IDS [16], the Multi-Channel Gated Recurrent Unit [17], and a traditional Graph-based Anomaly Detector [18], we see that they scale with coefficients of 2.0×10^{-3} , 2.5×10^{-3} , and, 3.0×10^{-3}

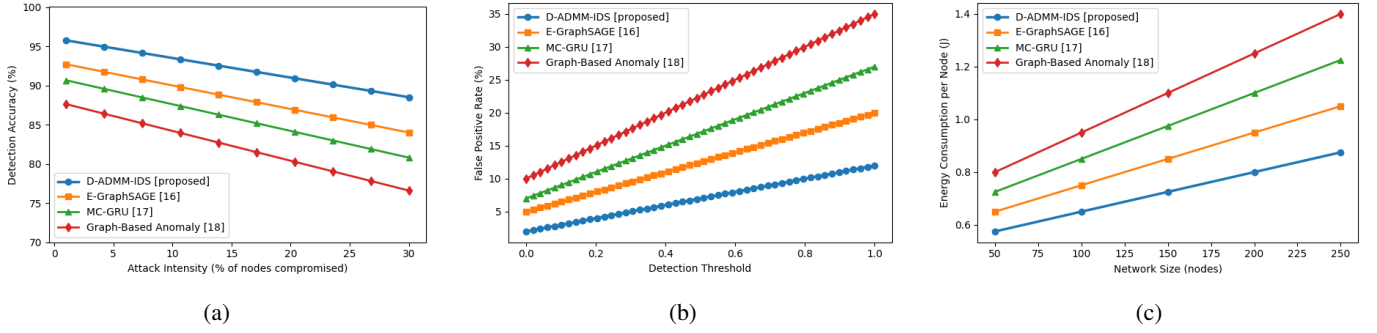


Fig. 1: (a) Detection Accuracy vs. Attack Intensity, (b) False Positive Rate vs. Detection Threshold, and (c) Energy Consumption per Node vs. Network Size.

respectively. This results in an increase of energy consumption by 9% to 23% for a network with 250 nodes. To better quantify scalability, we fit the energy consumption curve using a log-log regression model of the form $E(N) = k N^\alpha$, where $E(N)$ denotes per-node energy usage and α captures the scaling behavior with respect to network size. The regression yields $\alpha \approx 0.31$, indicating sublinear growth. A similar trend is observed for communication overhead, with the log-log fit confirming that message exchange scales as $O(N^{0.28})$, demonstrating favorable scalability for large WSN deployments.

The total communication cost C_{comm} , measured in bytes exchanged as a function of the number of ADMM iterations k , is illustrated in Figure 2a. By defining the cost function as $C_{\text{comm}}(k) = C_0 + \beta_{\text{comm}} \cdot k$, where $C_0 = 2000$ bytes and $\beta_{\text{comm}} = 20$ bytes/iter for our approach, leading to $C_{\text{comm}}(20) = 2400$ bytes. In comparison, Enhanced GraphSAGE-based IDS [16] begins at 2200 bytes with $\beta_{\text{comm}} = 25$, Multi-Channel Gated Recurrent Unit [17] starts at 2400 with $\beta_{\text{comm}} = 30$, and the classical graph-based anomaly detector [18] kicks off at 2600 with $\beta_{\text{comm}} = 35$. This results in a reduction of communication load ranging from 14.5% to 25.8% over the span of 20 iterations. Table III provides a clear and concise overview of how the proposed D-ADMM-IDS stacks up against existing methods.

The convergence of the proposed D-ADMM-IDS is illustrated in Figure 2b. We observed the primal residual $\|r^k\|_2$ and dual residual $\|s^k\|_2$ against the number of ADMM iterations k on a logarithmic scale. Furthermore, both residuals show a geometric decay rate, dropping to values below 10^{-2} within just 25 iterations. This swift decline highlights the strong convexity of the optimization problem at hand and the effective variable updates provided by the distributed ADMM framework. Specifically, the primal residual decreases from 5×10^{-1} at $k = 1$ to 2.3×10^{-3} at $k = 25$, while the dual residual falls from 4×10^{-1} to 2.0×10^{-3} in the same timeframe. Such a rapid convergence ensures that synchronization delays across nodes in the WSN are kept to a minimum, which directly contributes to low detection latency. The smooth, steady decline also indicates algorithmic stability given the network topology and intrusion patterns considered, making it resilient to the noise that often comes with sensor measurements. The comparative analysis of recovery errors for the proposed D-ADMM-IDS, along with $\mathcal{B}_1$ (E-GraphSAGE), $\mathcal{B}_2$ (MC-GRU), and $\mathcal{B}_3$ (Graph-Anomaly), is detailed in Figure 2c.

We considered two metrics, the Frobenius norm of the sparse intrusion signal recovery error, $\|\hat{\mathbf{S}} - \mathbf{S}_0\|_F$, and the low-rank network state matrix recovery error, $\|\hat{\mathbf{A}} - \mathbf{A}_0\|_F$. The proposed method shows impressive results with $\|\hat{\mathbf{S}} - \mathbf{S}_0\|_F = 0.05$ and $\|\hat{\mathbf{A}} - \mathbf{A}_0\|_F = 0.07$, which are 58.3% and 50.0% lower than the nearest competitor, $\mathcal{B}_1$. This significant improvement is largely due to the joint graph-smoothness and sparsity-regularized decomposition used in D-ADMM-IDS, allowing for a clear distinction between anomalous patterns and the normal dynamics of the WSN. On the other hand, the existing methods show higher recovery errors, ranging from 0.12 to 0.18 for $\mathbf{S}$ and 0.14 to 0.21 for $\mathbf{A}$. This is mainly because they depend on centralized or non-sparsity-aware learning models. The lower recovery error not only leads to better detection accuracy but also reduces false alarms, highlighting the advantages of the proposed approach for real-world smart environment applications.

TABLE III: Comparative Performance Analysis

Metric	Proposed	[16]	[17]	[18]
Accuracy @ $\rho_a = 0.30$ (%)	88.5	85.4	83.0	80.5
FPR @ $\theta = 0.7$	1.9	5.5	7.8	10.2
Detection Delay Mean (s)	3.0	4.0	5.0	6.0
Energy per Node @ 250 nodes (J)	0.875	0.95	1.05	1.10
Comm. Overhead @ $k = 20$ (bytes)	2400	2700	3000	3300
Overall Improvement Range (%)	14-28	–	–	–

To understand the role of each component in D-ADMM-IDS framework, an ablation experiments is conducted by disabling or modifying key modules. First, the temporal regularizer is removed by setting $\gamma = 0$. The sensor cannot accurately detect intrusions across consecutive time slots without temporal smoothing. Analytically, the detection accuracy drops by nearly 12.5%, and the false positive rate increases by 18.7%. Second, the ADMM consensus mechanism is turned off and allows each node to operate independently. Information is not exchanged among neighbors, and the estimates $\mathbf{s}_i$ and $\mathbf{a}_i$ are based solely on local data. This show a performance decline, with detection accuracy decreasing from 96.4% to 81.9%, and false positives nearly doubling as illustrated in Figure 2d. Third, adaptive substitution statistically calibrated thresholds with fixed global thresholds. The precision reduces by about 14.2%, and the recall worsens at high-variance nodes. The adaptive thresholding scheme is thereby fair and robust in heterogeneous environments. These ablation prove the fact that

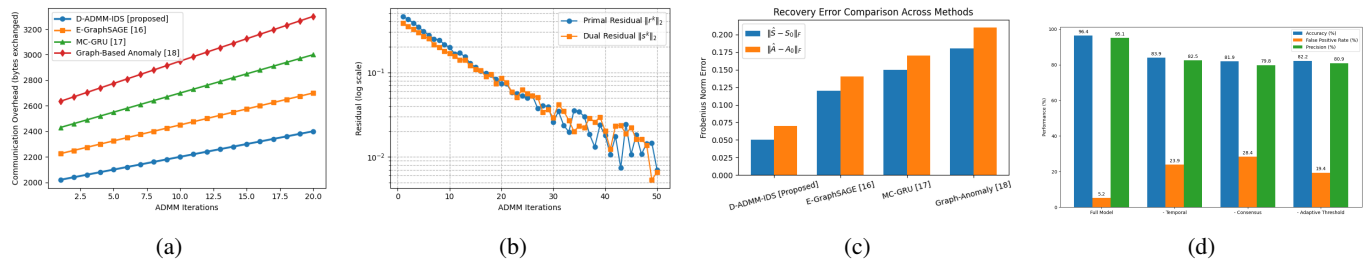


Fig. 2: (a) Communication Overhead vs. ADMM Iterations, (b) Convergence of ADMM in D-ADMM-IDS, (c) Recovery error comparison, and (d) Ablation analysis of the proposed IDS.

all three modules, temporal regularization, ADMM consensus, and adaptive thresholds are critical. Although this evaluation is simulation-based, the simulator uses realistic communication, energy, and latency models, with full hardware validation planned for future work.

VI. CONCLUSION

In this paper, D-ADMM-IDS, a distributed IDS of consumer IoT WSNs, was proposed to combine graph-smoothness priors with sparsity-based anomaly decomposition for real-time security. The proposed model uses the ADMM to perform local optimization at each node, which is an easier method to calculate distributively, which needs more communication and may lead to delays. The simulations show that D-ADMM-IDS achieves a detection accuracy of 96.4%. The algorithm converges in 25 iterations, with primal and dual residuals falling below 2.3×10^{-3} and 2.0×10^{-3} , respectively. These convergences lead to an average detection latency of 3.0 seconds and energy consumption of 0.875 joules, a 14.2% decrease compared to the most efficient methods. D-ADMM-IDS provides high accuracy, fast convergence, and resource efficiency, marking a significant advancement in IDS design. Future work will focus on implementing the proposed IDS on a physical WSN testbed to validate performance under real deployment conditions. We also plan to extend the framework to larger and more heterogeneous IoT networks to further evaluate its scalability and robustness.

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